

ChatGPT and Pre-Trained Language Models

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Abstracts

This report shows the application of AI-powered language models, particularly ChatGPT, in several fields, including education. ChatGPT's capabilities and constraints, and the potential advantages and disadvantages of adopting AI-based systems for teaching and learning are discussed.

Keywords: Chatgpt, AI, Chatbot, Pre-Trained, Neural Network, Deep learning, Machine learning, NLP

1. INTRODUCTION

Artificial intelligence helps humans by developing and analyzing sophisticated software and hardware, referred to as intelligent agents that can carry out a variety of tasks, AI has had an impact on how we go about our daily lives.

A chatbot is a computer program designed to simulate conversation with human users, especially over the Internet by the dictionary which is also taught by humans to a learning machine (Nuruzzaman and Hussain 2018). It communicates in human language via text or spoken communication with people or other chatbots using Natural Language Processing and sentiment analysis. Chatbots are sometimes referred to as interactive agents, smart bots, digital assistants, and artificial dialogue beings. The ability of chatbots to resemble human agents in communication is being improved by advances in AI. However, when compared to human-to-human communication, there are observable disparities in content and quality. A dialogue between a person and a chatbot lasts a long time. People frequently utilize brief sentences with little vocabulary, if not bad language. It is important to note that while chatbots are less capable of conversational understanding than humans are, the main distinction between them and people is how they perceive empathy. However, progress is being made, and chatbots are progressively becoming more sensitive to the emotions of those with whom they are interacting.

1.1. History

Alan Turing questioned if a computer program could communicate with a group of people in 1950 without the participants recognizing that their interlocutor was artificial. Many people believe this query, known as the Turing test, is where chatbots first started (Shahriar and Hayawi2 2020).

The primary purpose of Google Now, created in 2012, was to provide users with information while considering their preferences, location, and time of day. The successor to Google Now is Google Assistant, which was created in 2016. It offers information to consumers by anticipating their needs and has a more sophisticated artificial intelligence. The user interface is also friendlier and more conversational. However, because it is directly connected to the user's Google Account, it lacks personality, and its queries may breach their privacy.

Even though personal voice assistants provide voice connection with their users, misunderstandings frequently happen because these assistants are unable to grasp the specific language individuals use in oral speech or the overall context of the conversation (Adamopoulou and Moussiades 2020).

The use of chatbots has grown steadily, notably after 2016 (Fig. 1). The United States is the nation with the greatest amount of research interest in chatbots, while the United Kingdom and Japan are close behind with fewer than one-third as many articles published as in the USA (Fig. 2).

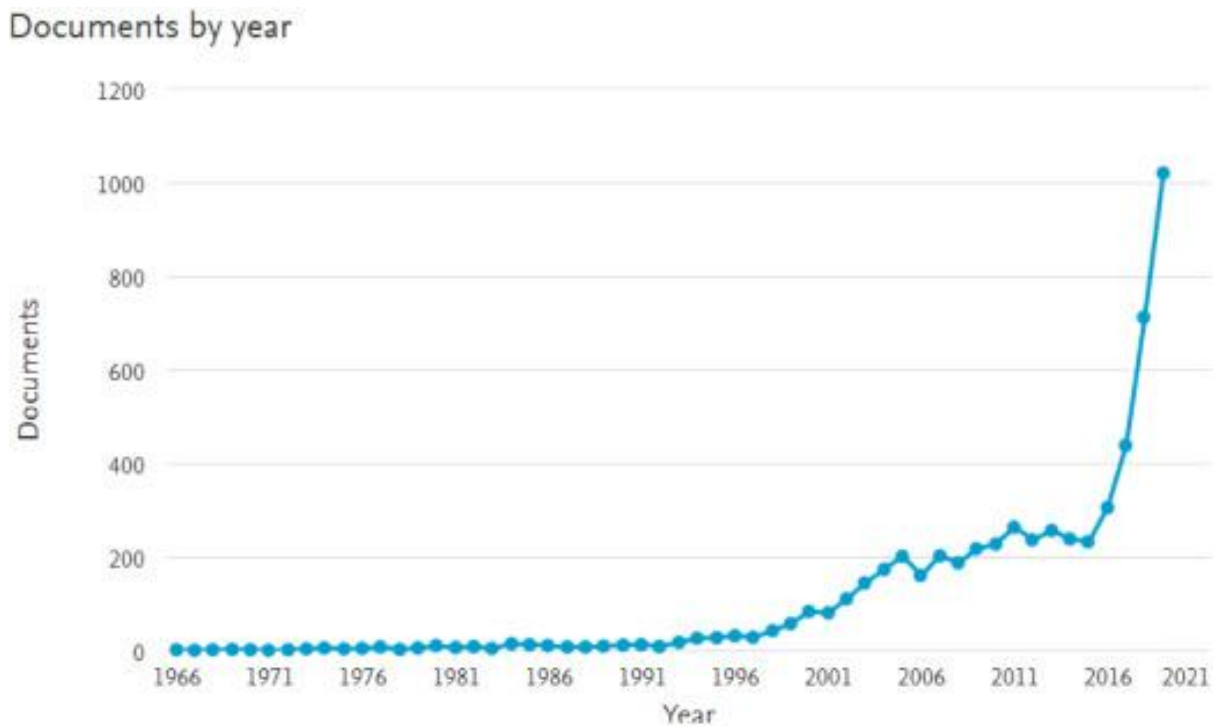


Figure.1: Results for the keywords "chatbot" "conversation agent" or "conversational interface" from 1966 to 2019
(Adamopoulou and Moussiades 2020).

Documents by country or territory

Compare the document counts for up to 15 countries/territories.

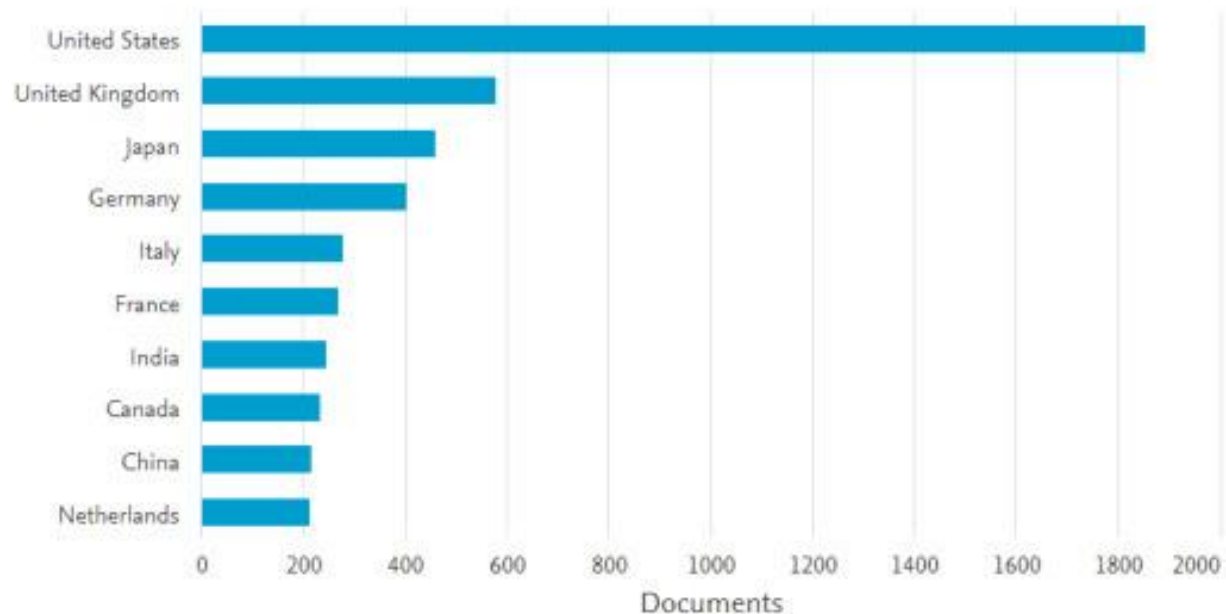


Figure.2: Shows the search results by country or territory for the terms "chatbot," "conversation agent," or "conversational interface" in Scopus from 1966 to 2019 (Adamopoulou and Moussiades 2020).

In today's digital world, chatbots are becoming more and more significant since they provide both organizations and individuals with several advantages. The following are some of the main benefits of chatbots (Adamopoulou and Moussiades 2020):

- **Better customer service:** Chatbots may offer round-the-clock customer support, instantly answer inquiries, and make personalized recommendations, which increases customer happiness and loyalty.
- **Cost-effective:** Because chatbots can manage numerous conversations at once, they cut down on the need for human customer service agents and consequently minimize the cost of customer support.
- **Timesaving:** Customers can save time and wait times by using chatbots to swiftly respond to frequently requested inquiries.
- **Increased engagement:** Chatbots make it simpler for businesses to interact with their audience by engaging with customers across numerous platforms, such as social media and messaging apps.
- **Data gathering:** By using chatbots to gather and analyze information on consumer behavior and preferences, businesses may enhance their goods and services.
- **Automation:** Chatbots can automate routine operations like scheduling appointments, reminding users of upcoming events, and processing orders, freeing up human workers' time to concentrate on more difficult jobs.

In general, the importance of chatbots resides in their capacity to improve customer engagement and satisfaction while streamlining operations and boosting efficiency.

Types of chatbots

A chatbot might be a member of more than one category at once. Chatbots are categorized based on a single simple criterion. A chatbot's knowledge domain is determined by the variety of knowledge it can access. Generic chatbots are those that can respond to any user inquiry from any domain. An illustration of a general chatbot is Chorus. Crosses or open-domain chatbots, like Guardian,

CRQA, or AskWiz, work across many domains. Domain-specific chatbots, however, like InstuctableCrowd, Legion: Mobile, or SnapTravel, can only react to inquiries about a specific knowledge domain.

Chatbot Categories	Knowledge domain	Generic
		Open Domain
		Closed Domain
	Service provided	Interpersonal
		Intrapersonal
		Inter-agent
	Goals	Informative
		Chat based/Conversational
		Task based
	Response Generation Method	Rule based
		Retrieval based
		Generative
	Human-aid	Human-mediated
		Autonomous
	Permissions	Open-source
		Commercial
	Communication channel	Text
		Voice
		Image

Figure.3: Classification of the chatbot(Adamopoulou and Moussiades 2020).

ChatGPT is a big language model built on the GPT-3 architecture that employs deep learning methods to respond to human-like text inputs (ALEX XU 2023). An overview of the architecture and procedure for producing answers is given in the diagram below:

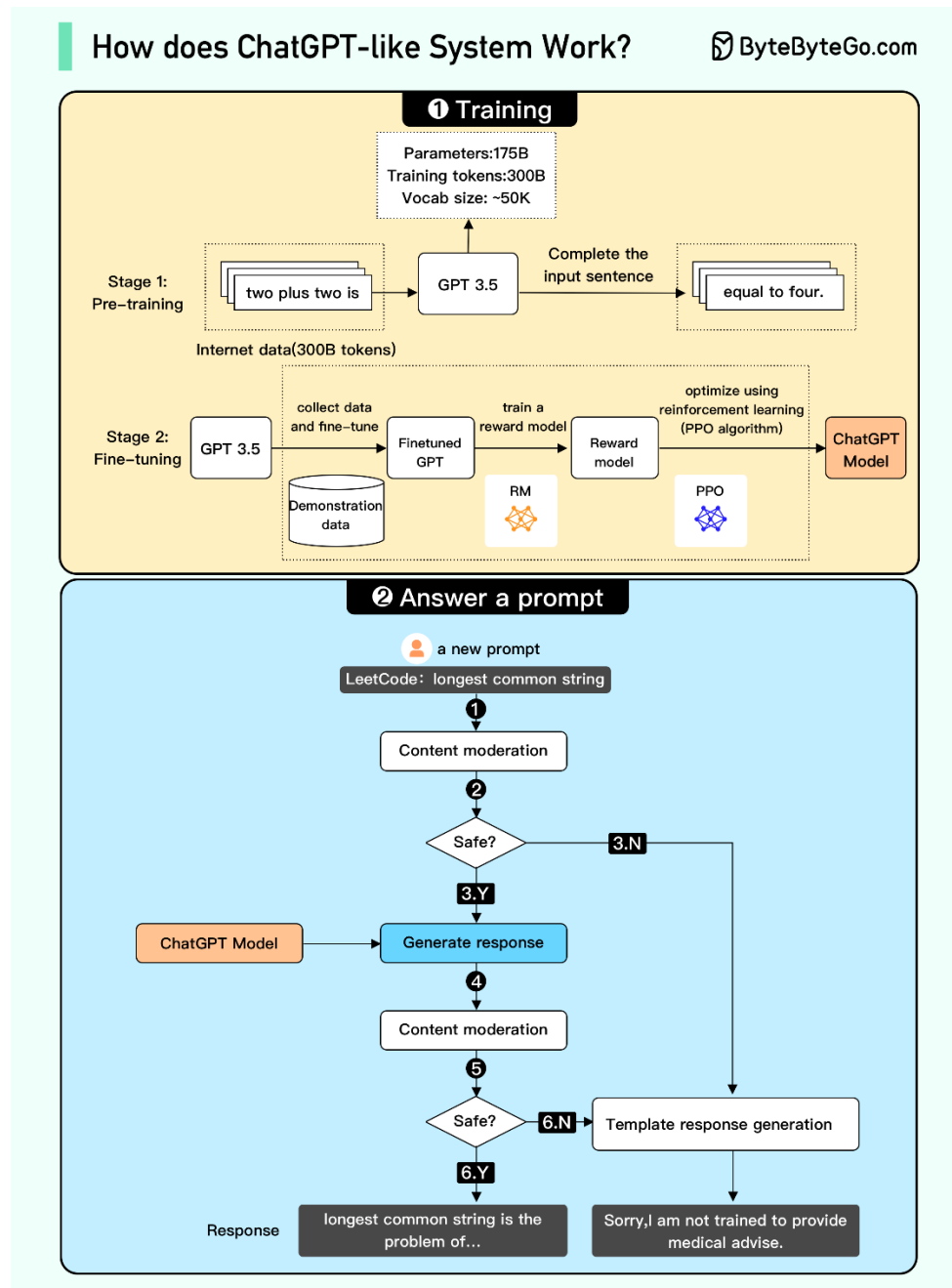


Figure.4: Architecture diagram for ChatGPT(ALEX XU 2023).

Here is a quick breakdown of each element (Brown et al. 2020):

1. **Text Input**: The user enters text into the system, which the system uses as the foundation for a response.
2. **Tokenization**: The input text is divided into separate tokens and supplied into the model in this process.
3. **Multi-Layer Transformer Model**: The Multi-Layer Transformer Model is the GPT architecture's primary component. The model consists of a few transformer layers, each of which transforms the input tokens and creates fresh tokens that are used by the levels below. The SoftMax function receives the output of the final layer and produces a probability distribution over the model's full vocabulary.

4. Sampling: The model chooses the subsequent token in the sequence to produce a response based on the probability distribution produced by the SoftMax function. Numerous sampling techniques are available, such as top-k sampling, nucleus sampling, and greedy sampling.
5. Output Text: After the sampling procedure is finished, the chosen tokens are concatenated to create a response, which is then sent back to the user.

2. LITERATURE REVIEW

The development of an AI-powered chatbot that can produce human-like phrases and compose well-organized essays has attracted the attention of the entire world (Shahriar and Hayawi2 2023). In addition, prospective ChatGPT uses in a range of industries including healthcare, education, and research are emphasized. Despite encouraging results, ChatGPT raises several privacy and moral problems.

Despite numerous drawbacks, artificial intelligence (AI) has proven useful and is being utilized more frequently in the field of radiology (Lecler et al. 2023). A language creation model created by OpenAI is called the Generative Pre-trained Transformer (GPT). GPT can produce language that sounds natural because it was trained on a big text corpus. It may be adjusted for a variety of language activities, including text completion, text summarization, and more. The most recent iteration of the device, GPT-3, is significantly more potent than its predecessors.

A GPT version designed primarily for comprehending and generating conversational language is called ChatGPT. It can comprehend the context and intent of a conversation because it was trained on a big collection of conversational text. In conversational settings, such as chatbots, virtual assistants, and other conversational interfaces, it can be used to generate human-like responses (Lecler et al. 2023).

Even if its talents in these areas still need development, as previously shown, ChatGPT has shown some unexpected abilities in the creation of computer code, journal entries, and financial statements (Street and Wilck 2023).

The entire globe has been interested in the creation of an AI-powered chatbot that can make human-like sentences and organize writing. Despite promising outcomes, ChatGPT poses several ethical and privacy issues (Shahriar and Hayawi2 2023).

ChatGPT and Related Generative AI's Benefits for Teaching and Learning. A very succinct (paragraph-length) and uninformative response is generated by ChatGPT.

Here are some examples of how ChatGPT can be used to improve learning, along with references to back them up (David Baidoo-Anu 2023):

- i. Personalized instruction: ChatGPT can be utilized to give students feedback and individualized instruction based on their unique learning requirements and advancement.
- ii. Automated Essay Grading: By being trained to grade student essays, ChatGPT can free up teachers' time to concentrate on other facets of the instruction.
- iii. ChatGPT can be used to translate instructional content into many languages, enhancing its accessibility to a wider audience.
- iv. Conversational Interaction with a Virtual Tutor: ChatGPT can be used to develop conversational interactions between students and a virtual tutor.

Using ChatGPT and other generative AI models in education has many potential advantages, but there are some disadvantages as well. Research findings back up these negatives:

- i. **Inadequate Human connection:** Unlike a genuine teacher or tutor, ChatGPT and other generative models cannot offer the same amount of human connection. Students who could gain more from a close relationship with a teacher may find it disadvantageous that there is little to no human interaction.
- ii. **Limited comprehension:** Because generative models are trained on statistical patterns in the data, they have a limited comprehension of the ideas that are being taught to pupils. When it comes to giving explanations or feedback that are specific to a student's requirements and misconceptions, this may be a drawback.
- iii. **Lack of Originality:** Generative models may only provide responses based on the patterns in the data they have observed during training, which can restrict their ability to produce solutions that are both creative and unique.
- iv. **Data Dependency:** Generative models require a lot of data to train, and both the quality and quantity of the data have a significant impact on the model's performance. The model won't function as well if the data is insufficient or irrelevant.
- v. **Contextual Understanding Deficit:** Generative models are unable to comprehend the circumstance and context, which can result in responses that are inappropriate or irrelevant.
- vi. **Limited ability to tailor training:** While ChatGPT and other generative AI models can offer broad information and help, they may be unable to tailor instruction to a specific student's needs.
- vii. **Privacy:** Using ChatGPT and other generative AI models in education raises questions regarding data security and privacy. It's critical to remember that while ChatGPT and other generative AI models are effective tools, they cannot take the role of human instructors and tutors. It's crucial to utilize these technologies sensibly and in conjunction with guidance and assistance from others.

While generative AI models like ChatGPT can be effective tools for improving teaching and learning, it's crucial to be aware of their limitations and to utilize them in a way that respects those limitations together with other teaching strategies that emphasize communication and comprehension between people.

Along with DALL-E-2 and Codex, ChatGPT has recently attracted a lot of public attention. As a result, a lot of people have developed an interest in connected resources and are trying to learn more about the history and techniques that led to its remarkable performance. ChatGPT and other generative AI methods fall under the umbrella of artificial intelligence-generated content (AIGC), which is the process of using AI models to generate digital material including music, images, and spoken language (Cao et al. 2023). The objective of AIGC is to increase the accessibility and efficiency of the content creation process so that high-quality content may be produced more quickly.

Pros and Cons of AI-based ChatGPT. The combined pros and cons of using ChatGPT inside regular classrooms (Karthikeyan and Karthikeyan 2022):

Pros:

ChatGPT has faster informational access to all users. It encourages individualized instruction and can research a greater range of subjects all at once with an additional resource. By requesting instant feedback, it can aid in a speedy search than most of the searching methods. ChatGPT can have accessibility around-the-clock and compete with teachers' pace.

Cons:

ChatGPT has limited accuracy because there is only one possible response. Also, lack of context will be a problem. It could generate inaccurate output if the input data is incorrect. There won't be any critical thinking from the prevention of lateral thought. Ingenuity won't be present. The one-sidedness of the responses. Purely dependent on technology, and in the absence of technology, efficiency decreases.

When assisting a scientist in producing a research article, the main drawbacks of GPT-type models like ChatGPT are that they lack subject expertise, depending on the quality of the input data, are unable to check for errors, lack creativity, and lack an awareness of ethical problems. GPT-like models can produce content that is coherent and grammatically accurate, but it could lack the nuances and context that a human specialist in a particular field, like radiology, would have. The results produced could not be pertinent or it could not be accurate (Lecler et al. 2023).

OpenAI has already tested a Deep Neural Network (DNN) utilizing a substantial sentence library, and it has improved its functionality for tasks like producing phrases or replying to queries (Gupta et al. 2023).

Chat GPT-3 is a significant advancement in Natural Language Processing (NLP), which makes use of a transformer-built structure to analyze large amounts of data concurrently and produce a language that is more like what a human would understand. This breakthrough has a variety of uses, such as text classification services, bots, and machine translation software (Gupta et al. 2023). However, Chat GPT-3 is unable to connect to the Web and is only able to use the Internet it has grown accustomed to, which limits its capacity to learn new information. The word cloud contains phrases associated with artificial intelligence, such as "Machine Learning," "ChatGPT," "Neural Networks," and "Deep Learning." The phrases "natural language processing," "language generation," and "text completion" are also included.

There will always be a need for consumers to connect with banks for financial services, and because of these interactions, data is produced that are advantageous to both sides. There is a chance for ChatGPT to investigate these datasets and use the results to guide some banking decisions. Since banks are already urging clients to "come bank with them," they must demonstrate some comprehension, commitment, and awareness of those customers' demands. For their front-end operations, banks can employ ChatGPT technology to improve business processes and interact with customers. The chatbot, which has been successfully incorporated into banking operations and is utilized to improve customer experiences by answering commonly asked questions and streamlining the banking connection, is built upon by this conversation agent. Banks can improve client services, build better algorithms, and obtain information about the quality of their service delivery. However, as ChatGPT makes its way into banking operations, customer confidence in the quality of the service will be crucial. Consumers doubting their confidence in the offered services and information would not be unexpected. Banks will be expected to invest in training employees and informing clients about the options accessible to them. Banks must understand their customers' wants and how far to push the technology drive because many customers still need to be able to communicate with chatbots.

These are the graph outlining three stages of implementation of ChatGPT: -

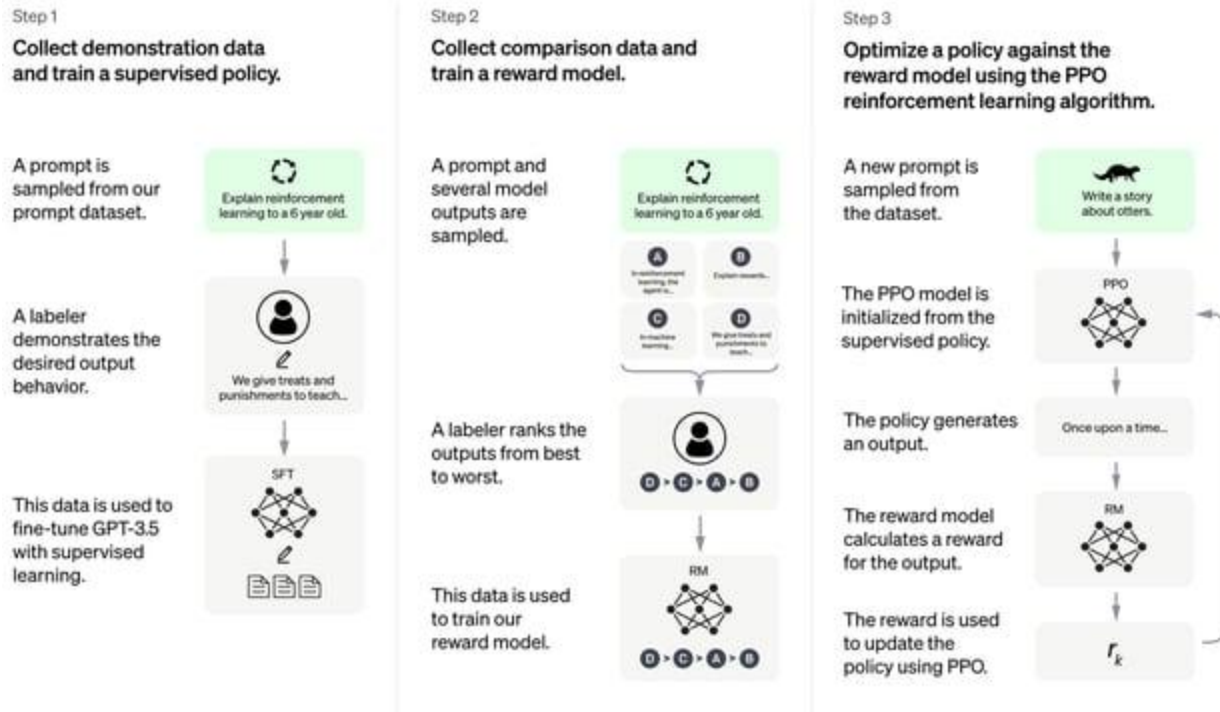


Fig.5. Technical implementation of Chatgpt (HPC-AI Tech 2023).

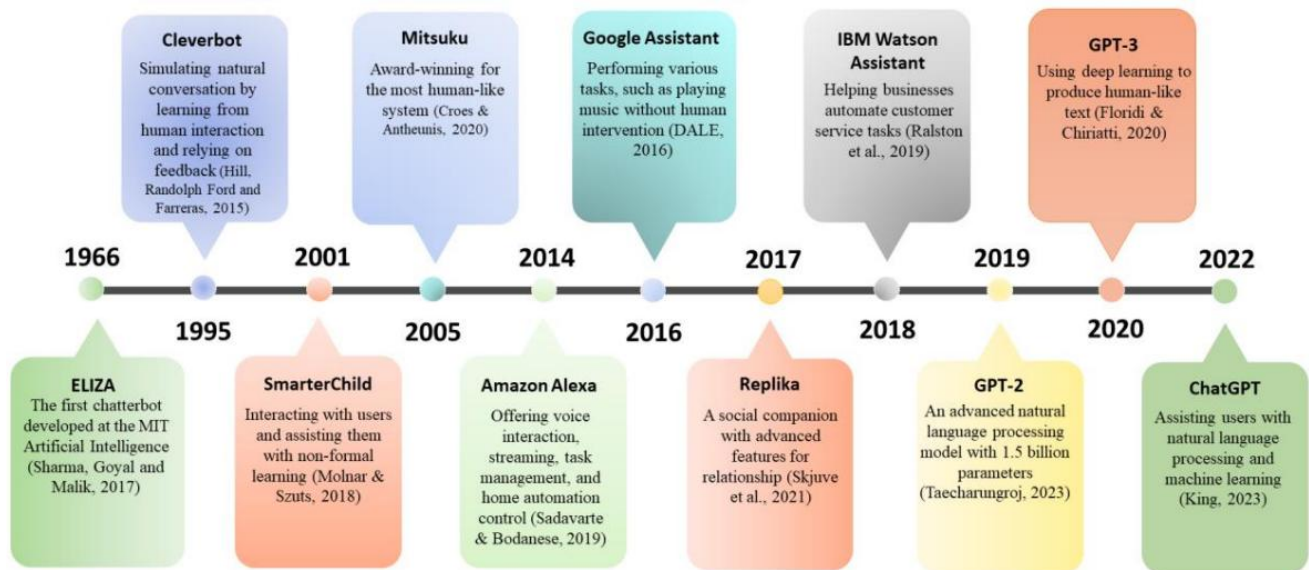


Fig.6. Some well-known chatbots throughout history (Khosravi et al. 2023).

Due to their adaptability in fields including healthcare, customer service, and education, chatbots have risen in popularity in recent years (Khosravi et al. 2023). The GPT-3.5 architecture-based ChatGPT, created by OpenAI, stands out among these chatbots for its sophisticated natural language processing capabilities. ChatGPT enables users to ask questions, give answers, and have discussions on a range of subjects that are pertinent to various fields. scholars from almost 60% of the world's countries and territories have participated in chatbot research, which has attracted interest from scholars throughout the globe. Despite being a recent subject, research on ChatGPT has been carried out in 46 nations.

With literature mostly focusing on its potential and limits in fields like research ethics, medicine, and social science, ChatGPT is seen as a promising research direction in chatbot research. The study discovered that ChatGPT has the potential for a wide range of applications, including answering medical questions, creating clinical decision and support systems, improving discharge summaries, assisting with writing, translating, simulating interactions between employees and organizations, and supporting policy decision-making, despite some limitations and the need for human assistance at times.

Before ChatGPT is widely adopted, research on its effects from a healthcare viewpoint in a practical environment should be done (Sallam 2023). To mitigate the potential negative effects of any potential abuse, the real-world effects of ChatGPT in comparison to other LLMs should be thoroughly assessed. The same ground-breaking tool might be extremely harmful if applied incorrectly. A case study based on Formula 1 racing can be used to demonstrate the extremely detrimental effects of ChatGPT abuse as follows. When Michael Schumacher, one of the most successful Formula 1 drivers of all time, was behind the wheel of the Ferrari F2004 during the 2004 Formula 1 season, the car smashed a few F1 records. The same is true for the extremely successful car, however, would only destroy walls and be irreparably wrecked in my hands as a modest researcher without experience in Formula 1 driving. However, ChatGPT's capability has the potential to hurt by providing false or erroneous content, raising worries about scholarly disinformation. The potential dissemination of false information in medical publishing can pose serious societal risks, as the so-called COVID-19 infodemic demonstrates. ChatGPT occasionally produces replies that are true or illogical but plausible-sounding.

3 ANALYSIS

3.1 Ethical issues

Although, the amazing powers of big language models, such as GPT-3, can pick up new jobs with little guidance. ChatGPT as an AI language model brings up several ethical concerns that have been hotly debated in the AI community (Anna Jobin 2019). ChatGPT may have access to sensitive user data, such as contact details and discussion logs. Making sure that this data is secure and only used for legitimate purposes is crucial. People and society may suffer significantly if AI systems like ChatGPT misuse user data. The data that ChatGPT was trained on may have contained prejudices and discriminatory views, which may be reflected in its responses (Gupta EPFL and Jaggi EPFL 2021). ChatGPT has restrictions on innovation and diversity in NLP and has moral ramifications (Lei et al. 2021). As a result, certain groups of people may receive biased treatment depending on their color, gender, or other traits. Large language models like GPT-3, according to studies, frequently reinforce biases that are already present in the data they are trained on. Openness can lessen problems with prejudice and justice in AI systems. Policymakers and stakeholders should be considered for any potential ethical ramifications of AI systems (Zuo et al. 2019).

3.2 Trust issues

Given the model's opaque decision-making process and the potential for biased or improper responses, users have serious concerns about ChatGPT's trust difficulties (Mittelstadt 2019). ChatGPT's decision-making process is frequently opaque, making it challenging to comprehend how it determines its responses. For users to have confidence in the decisions made by AI systems like ChatGPT, it is crucial to make sure that they are clear and explicable. The theoretical foundation for creating trust in chatbots should be based on three essential characteristics: competence, goodness, and integrity. Chatbots lack several design techniques for fostering user confidence in chatbots, such as giving clear feedback and justifications for system choices (Nur AL-huda Hamdan 2019). AI systems' poor transparency and explain ability is a significant barrier to their adoption in crucial industries like finance and healthcare. Users are worried about privacy and trust when using intelligent personal assistants (IPAs) due to the imbalance of information between users and IPAs including Siri and Alexa (Syed Muhammad Umair Arif 2019). Design solutions should be done with this asymmetry, including having transparent privacy policies and enabling users to view and remove their data manually could help to improve the issues caused by chatbots. Chatbot users can better comprehend and have faith in these systems if they are transparent, or open about how they make decisions (Fabrice Matulic 2020).

3.3 Accountability issue

ChatGPT lacks duty and accountability as an AI system since it lacks a conscience (Russell 2020). However, it is the responsibility of those who create and implement the system to guarantee that it functions morally and in the interests of society. This necessitates a commitment to ethical standards like fairness, openness, and responsibility as well as a thorough knowledge of the possible risks and advantages of AI systems like ChatGPT. Transparency is essential for

fostering accountability and moral AI practices (Wu et al. 2018). Transparency can lessen problems with explainability in AI systems (Wu et al. 2018). There are potential risks connected to the creation and application of AI systems for tactical advantage, as well as the necessity of openness and cooperation to lessen these risks (Zuo et al. 2019).

4. CONCLUSION

With the remarkable performance of ChatGPT on a variety of linguistic problems, it is a potent tool for natural language processing. The model still has drawbacks and difficulties, such as the possibility of producing poisonous or prejudiced language and the need for a huge amount of training data. To fully understand ChatGPT and other pre-trained language models in a variety of scenarios, including chatbot generation, style transfer, and text classification, more study is required. When creating and implementing language models like ChatGPT, ethical issues like bias and fairness must be considered. Even though ChatGPT has the potential to significantly advance the field of natural language processing, researchers and developers must continue to carefully assess and enhance the model to guarantee its ethical and efficient application.

I believe that ChatGPT's usability and accessibility could significantly improve scholarly output if the program is made publicly available soon. Since ChatGPT can receive and produce content in a variety of languages, it could democratize the sharing of information by removing the need for English-language constraints that can prevent speakers of other languages from publishing their work. The early rollout of ChatGPT will inevitably spawn competitors, potentially rendering this a far-reaching problem. The moment for these interactions is now, as shown by ChatGPTs.

5. REFERENCES

- Adamopoulou, E. and Moussiades, L. 2020. Chatbots: History, technology, and applications. *Machine Learning with Applications* 2, p. 100006. doi: 10.1016/j.mlwa.2020.100006.
- ALEX XU. 2023. EP 44: How does ChatGPT work?
- Anna Jobin, M.I.& E.V. 2019. The global landscape of AI ethics guidelines. *The global landscape of AI ethics guidelines*.
- Brown, T.B. et al. 2020. Language Models are Few-Shot Learners. Available at: <http://arxiv.org/abs/2005.14165>.
- Cao, Y., Li, S., Liu, Y., Yan, Z., Dai, Y., Yu, P.S. and Sun, L. 2023. A Comprehensive Survey of AI-Generated Content (AIGC): A History of Generative AI from GAN to ChatGPT. Available at: <http://arxiv.org/abs/2303.04226>.
- David Baidoo-Anu, L.O.A. 2023. Education in the Era of Generative Artificial Intelligence. *Education in the Era of Generative Artificial Intelligence (AI): Understanding the Potential Benefits of ChatGPT in Promoting Teaching and Learning*.
- Fabrice Matulic, R.A.B.V.D.V. 2020. PenSight: Enhanced Interaction with a Pen-Top Camera. *PenSight: Enhanced Interaction with a Pen-Top Camera*.
- Gupta, B., Mufti, T., Saquib Sohail, S. and Øivind Madsen, D. 2023. ChatGPT: A Brief Narrative Review. Available at: www.preprints.org.
- Gupta EPFL, P. and Jaggi EPFL, M. 2021. *Obtaining Better Static Word Embeddings Using Contextual Embedding Models*. Available at: <https://github.com/epfml/X2Static>.

HPC-AI Tech. 2023. Replicate ChatGPT Training Quickly and Affordable with Open Source Colossal-AI. *Replicate ChatGPT Training Quickly and Affordable with Open Source Colossal-AI*.

Karthikeyan, K.C. and Karthikeyan, C. 2022. Literature Review on Pros and Cons of ChatGPT Implications in Education View project Research Based Book on Crowdfunding View project Literature Review on Pros and Cons of ChatGPT Implications in Education. *International Journal of Science and Research*. Available at: www.ijsr.net.

Khosravi, H., Shafie, M.R., Hajiabadi, M., Raihan, A.S. and Ahmed, I. 2023. *Chatbots and ChatGPT: A Bibliometric Analysis and Systematic Review of Publications in Web of Science and Scopus Databases*.

Lecler, A., Duron, L. and Soyer, P. 2023. Revolutionizing radiology with GPT-based models: Current applications, future possibilities and limitations of ChatGPT. *Diagnostic and Interventional Imaging*. doi: 10.1016/J.DIII.2023.02.003.

Lei, J., Li, L., Zhou, L., Gan, Z., Berg, T.L., Bansal, M. and Liu, J. 2021. Less is More: ClipBERT for Video-and-Language Learning via Sparse Sampling. Available at: <http://arxiv.org/abs/2102.06183>.

Mittelstadt. 2019. The ethics of algorithms: Mapping the debate. Big Data & Society. *The ethics of algorithms: Mapping the debate*. Big Data & Society.

Nur AL-huda Hamdan, A.W.S.V.J.S.J.B. 2019. Springlets: Expressive, Flexible and Silent On-Skin Tactile Interfaces. *Springlets: Expressive, Flexible and Silent On-Skin Tactile Interfaces*.

Nuruzzaman, M. and Hussain, O.K. 2018. A Survey on Chatbot Implementation in Customer Service Industry through Deep Neural Networks. In: *Proceedings - 2018 IEEE 15th International Conference on e-Business Engineering, ICEBE 2018*. Institute of Electrical and Electronics Engineers Inc., pp. 54–61. doi: 10.1109/ICEBE.2018.00019.

Russell. 2020. Artificial intelligence: A modern approach. *Artificial intelligence: A modern approach*.

Sallam, M. 2023. ChatGPT Utility in Healthcare Education, Research, and Practice: Systematic Review on the Promising Perspectives and Valid Concerns. *Healthcare (Basel, Switzerland)* 11(6). Available at: <http://www.ncbi.nlm.nih.gov/pubmed/36981544>.

Shahriar, S. and Hayawi², K. 2020. *Let's have a chat! A Conversation with ChatGPT: Technology, Applications, and Limitations*. Available at: <https://www.forbes.com/sites/ariannajohnson/2022/12/07/heres-what-to-know-about-openais-chatgpt-what-its->.

Shahriar, S. and Hayawi², K. 2023. *Let's have a chat! A Conversation with ChatGPT: Technology, Applications, and Limitations*. Available at: <https://www.forbes.com/sites/ariannajohnson/2022/12/07/heres-what-to-know-about-openais-chatgpt-what-its->.

Street, D.A. and Wilck, J. 2023. *“Let's Have a Chat”: Applying ChatGPT and other large language models to the practice of forensic accounting*. Available at: <https://openai.com/blog/chatgpt/>.

Syed Muhammad Umair Arif, P.M.F.B. 2019. A Comparative Study of Rendering Devices for Safety-Critical Applications in Operative Control Rooms. *A Comparative Study of Rendering Devices for Safety-Critical Applications in Operative Control Rooms*.

Wu, F., Hwang, E. and Sarma, S. Das. 2018. Phonon-induced giant linear-in- T resistivity in magic angle twisted bilayer graphene: Ordinary strangeness and exotic superconductivity. Available at: <http://arxiv.org/abs/1811.04920>.

Zuo, Y., Jin, S. and Zhang, S. 2019. Computation Offloading in the Untrusted MEC-aided Mobile Blockchain IoT System. Available at: <http://arxiv.org/abs/1911.08255>.